

GEORGE W. ANDREWS PERTINENT DATA

GENERAL

Damsite, miles above mouth of Chattahoochee River	46.5
Drainage area above damsite – square miles	8,210
Drainage area between damsite and Walter F. George Dam – square miles	750

RESERVOIR

Length – river miles	28.3
Minimum operating pool elevation, feet NGVD29	96.0
Normal operating pool elevation, feet, NGVD29	102.0
Maximum operating pool elevation, feet NGVD29	103.0
Peak flood for standard project flood, feet, NGVD29	130.2
Peak pool for spillway design flood, feet, NGVD29	135.5
Area at static full pool (elev. 102) – acres	1,540
Total volume at static full pool (elev. 102) – acre-feet	18,180
Shore line length at static full pool – miles	65

TAILWATER ELEVATIONS (feet above NGVD29)

Highest observed elevation at site (18 Mar 1929, peak flow – 203,000 cfs)	125.8
Capacity at headwater elevation 102 (43,600 cfs)	100.7
Capacity at headwater elevation 96 (30,800 cfs)	94.5
Bankfull capacity for downstream reach (40,000 cfs, approx.)	99.7
Minimum tailwater (at zero discharge)	76.0

DAM AND SPILLWAY

Total length – feet	620
Fixed-crest length – feet	340
Fixed-crest elevation – feet, NGVD29	102.0
Type of gates	Tainter
Size of gates – feet	60 x 21
Top of gate elevation (closed) – feet, NGVD29	103
Number of gates	4

LOCK

Static full upper pool elevation – feet, NGVD29	102
Normal lower pool elevation – feet, NGVD29	77
Maximum lift – feet	25
Chamber nominal length x width – feet	450 x 82
Elevation of top of lock walls – feet, NGVD29	115
Elevation top of upper sill – feet, NGVD29	83
Elevation of top of lower sill – feet, NGVD29	63
Elevation of floor – feet, NGVD29	60
Type of gates	Miter

STREAM FLOW AT DAMSITE

Average annual flow, (May 1938 – Sep 2013) – cfs	10,767
Minimum mean daily, 30 Jan 1989 (May 1938 – Sep 2013) – cfs	498
Maximum mean daily, 8 July 1994 (May 1938 – Sep 2013) – cfs	195,000
Maximum known, 18 Mar 1929 – cfs	203,000
Bankfull (approx.) for downstream reach – cfs	40,000
Minimum mean monthly flow, Oct 1954 – cfs	1,376
Maximum mean monthly flow, Mar 1990 (1938 – 2009) – cfs	45,879
Maximum known mean monthly flow, Mar 1929 – cfs	99,600
Standard project flood peak inflow – cfs	293,100
Spillway design flood peak inflow – cfs	630,000

TAILWATER ELEVATIONS (feet above NGVD29)

Highest observed elevation at site (18 Mar 1929, peak flow – 203,000 cfs)	125.8
Selected spillway discharges:	
Capacity at headwater elevation 102 (43,600 cfs)	100.7
Bankfull capacity for downstream reach (40,000 cfs, approx.)	99.7
Capacity at headwater elevation 96 (30,800 cfs)	94.5
Full Walter F. George power operation, best gate, no reregulation (25,000 cfs, approx.)	92.5
Peaking operation Walter F. George power plant:	
Reregulated peak discharge, 20,000 cfs	87.9
Reregulated peak discharge 15,000 cfs	84.8
Reregulated peak discharge 10,000 cfs	81.8
Minimum tailwater (zero discharge)	76.0

DAM AND SPILLWAY

Total length – feet	620
Fixed-crest length – feet	340
Fixed-crest elevation – feet, NGVD29	102.0
Gated length – feet	280
Gated crest elevation – feet, NGVD29	82
Gated net length – feet	240
Number of piers	5
Width of piers – feet	8
Service bridge elevation – feet, NGVD29	142
Type of gates	Tainter
Size of gates – feet	60 x 21
Top of gate elevation (closed) – feet, NGVD29	103
Number of gates	4
Capacity of gated spillway at pool elevation 102 – cfs	43,600
Capacity of gates spillway at pool elevation 96 – cfs	30,800
Apron Elevation	Varies
Apron length below gated section – feet	34

LOCK

Static full upper pool elevation – feet, NGVD29	102
Normal lower pool elevation – feet, NGVD29	77
Maximum lift – feet	25
Chamber width – feet	82
Nominal chamber length – feet	450
Distance between pintles – feet	505
Elevation of top of lock walls – feet, NGVD29	115
Elevation top of upper sill – feet, NGVD29	83
Elevation of top of lower sill – feet, NGVD29	63
Elevation of floor – feet, NGVD29	60
Type of gates	Miter
Height of lower gate – feet	51
Height of upper gate – feet	32
Types of valves	Tainter
Culvert size – feet	10 x 10
Freeboard when lock becomes inoperative – feet	1
Percent of time inoperative	58